Claude's sessions: RT by trial, transitivity and consistency

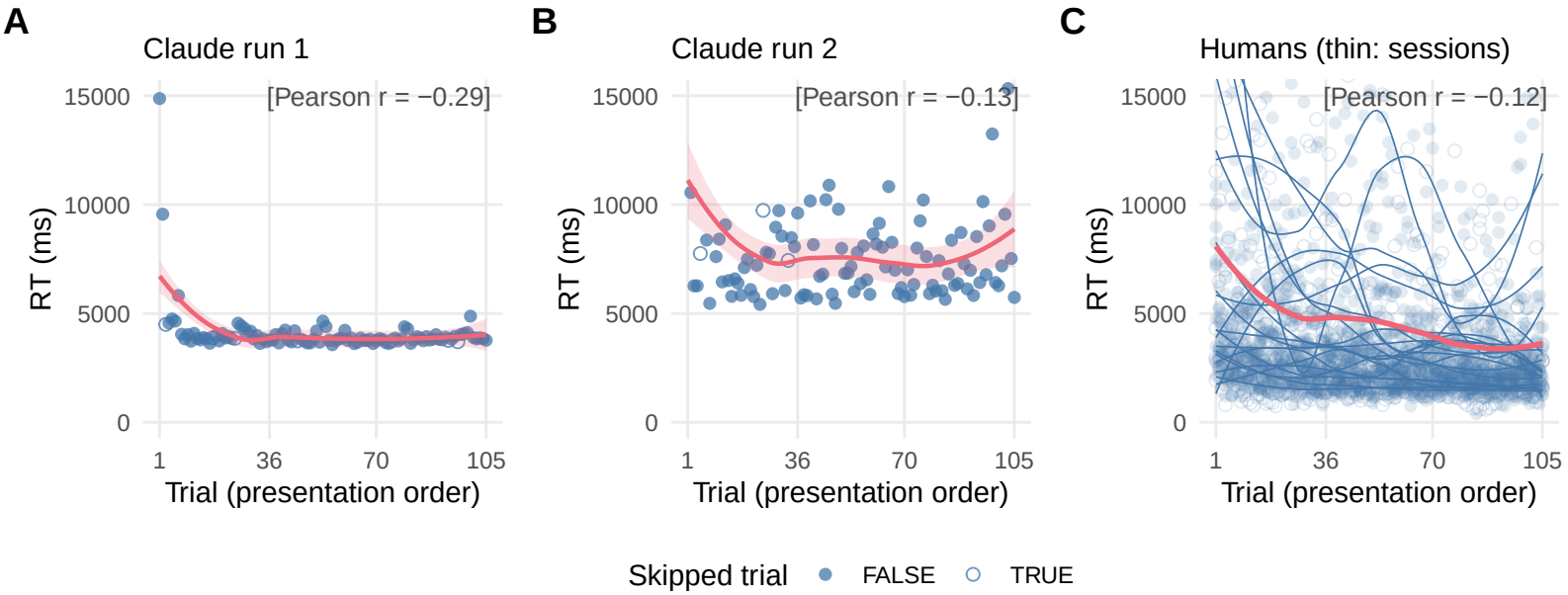
Data. Claude's raw session CSVs (read directly, bypassing preprocessing) and the human pilot reference from data/processed/cbcu.csv (10 participants x 2 sessions x 105 trials).

Claude run 1 (PHQ9_pairwise_kpd7ta7l_SESSION_2026−10−09_11h00.23.625.csv): 242 raw rows, 105 choice rows kept
Claude run 2 (PHQ9_pairwise_spf06mb2_SESSION_2026−10−12_10h23.05.926.csv): 242 raw rows, 105 choice rows kept
Only rows with trial_type html−button−response and phase pairwise are kept; all other rows are dropped (row types of run 1 shown; run 2 has the same structure):

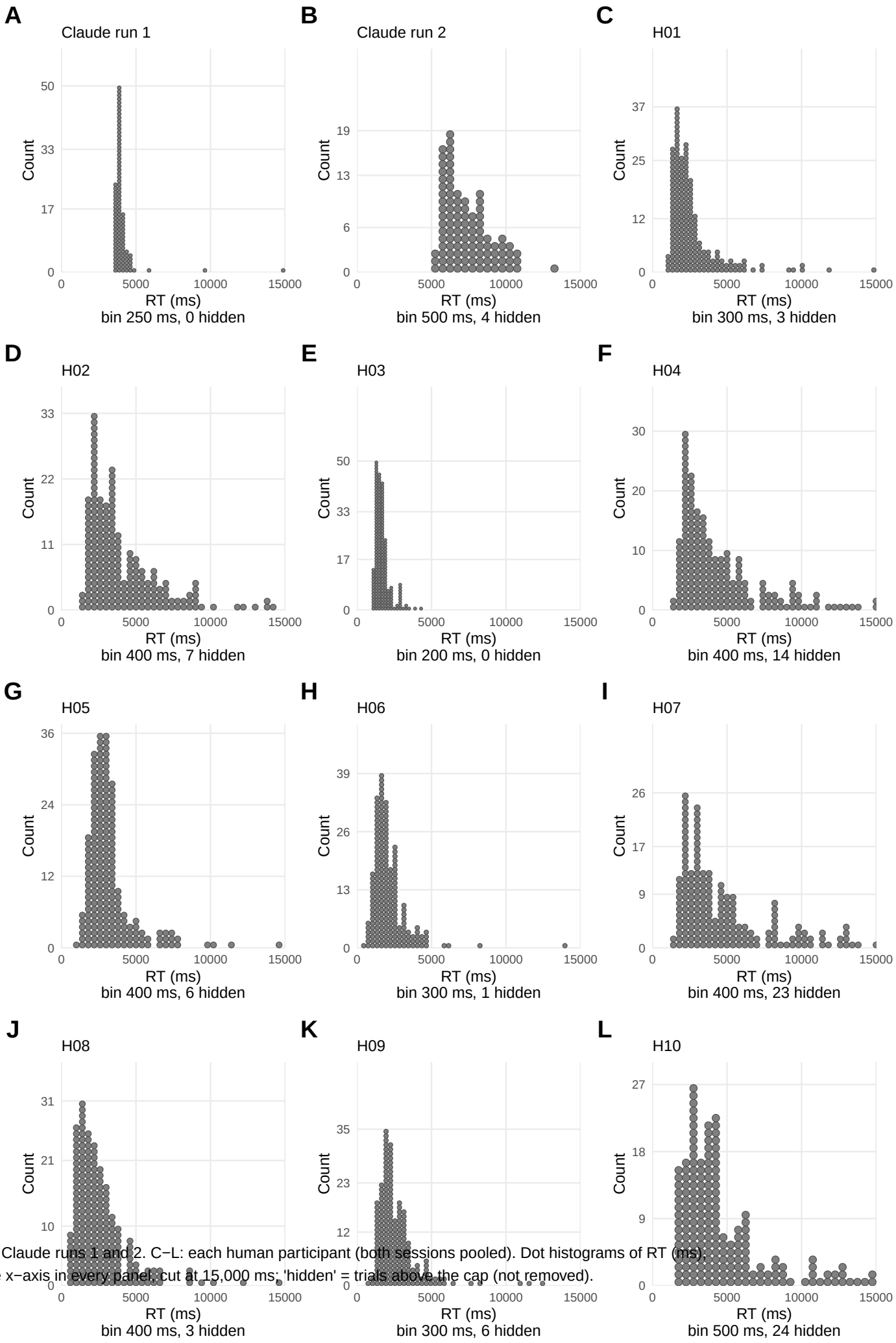
trial_type	phase	rows	kept
fullscreen	(none)	1	dropped
html−button−response	pairwise	105	KEPT
html−button−response	phase2_break	3	dropped
html−button−response	phase2_instructions	7	dropped
html−button−response	phase2_ready	1	dropped
html−button−response	phq9_grid	1	dropped
html−button−response	quiz	6	dropped
html−button−response	(none)	7	dropped
html−keyboard−response	iti	105	dropped
html−keyboard−response	(none)	3	dropped
pavlovia	(none)	1	dropped
survey−text	feedback	1	dropped
survey−text	free_text_explanation	1	dropped

Checks that passed (the script stops if any fails): exactly 105 choice rows; trial order 1−105 without duplicates; no repeated pair; no missing RT; window_status 'ok' on every row; choice values left/right/skip; skip matches the skipped flag; chosen item matches the chosen side. No RT outliers were removed.

Human reference: 20 participant−sessions, 105 trials each (checked).
RT axis cap in the figure: 15,000 ms (trials above it are hidden, not removed):
Claude: 4 of 210 trials above 15000 ms
Human: 87 of 2100 trials above 15000 ms



A, B: Claude runs 1 and 2, RT (ms) against presentation order, LOESS line with band; hollow points are skipped trials.
C: human sessions, one thin LOESS line per session, pooled line in rose.



A, B: Claude runs 1 and 2. C–L: each human participant (both sessions pooled). Dot histograms of RT (ms), same x-axis in every panel, cut at 15,000 ms; 'hidden' = trials above the cap (not removed).

Transitivity and consistency

Skip trials are ignored for transitivity and consistency. A cycle is A over B, B over C and C over A. cycle_rate = cyclic triples / triples with all three pairs decided; kendall_zeta = 1 − cycle_rate / 0.308 (1 = perfectly transitive). contradiction_share = fewest choices contradicting any single ranking / decided trials (0 = one ranking explains every choice). Participants with many skips have few decided triples, so their rates are unstable; H08 skipped every trial.

unit	n_decided	skip_rate	n_decided_triples	n_cyclic	cycle_rate	kendall_zeta	min_contradictions	contradiction_share
Claude_run1	98	0.067	375	0	0.000	1.000	0	0.000
Claude_run2	102	0.029	418	0	0.000	1.000	0	0.000
H01_time1	99	0.057	385	1	0.003	0.992	1	0.010
H01_time2	99	0.057	385	0	0.000	1.000	0	0.000
H02_time1	104	0.010	442	10	0.023	0.926	4	0.038
H02_time2	104	0.010	442	10	0.023	0.926	4	0.038
H03_time1	26	0.752	12	0	0.000	1.000	0	0.000
H03_time2	27	0.743	13	0	0.000	1.000	0	0.000
H04_time1	85	0.190	260	14	0.054	0.825	4	0.047
H04_time2	92	0.124	318	13	0.041	0.867	3	0.033
H05_time1	76	0.276	198	1	0.005	0.984	1	0.013
H05_time2	70	0.333	161	6	0.037	0.879	2	0.029
H06_time1	103	0.019	430	21	0.049	0.841	6	0.058
H06_time2	105	0.000	455	62	0.136	0.557	13	0.124
H07_time1	28	0.733	17	0	0.000	1.000	0	0.000
H07_time2	36	0.657	32	1	0.031	0.898	1	0.028
H08_time1	0	1.000	0	0	NA	NA	0	NA
H08_time2	0	1.000	0	0	NA	NA	0	NA
H09_time1	86	0.181	269	0	0.000	1.000	0	0.000
H09_time2	81	0.229	232	16	0.069	0.776	5	0.062
H10_time1	104	0.010	442	15	0.034	0.890	6	0.058
H10_time2	105	0.000	455	18	0.040	0.871	7	0.067

Human range (min / median / max) next to Claude

measure	min	median	max	Claude_run1	Claude_run2
skip_rate	0.000	0.186	1.000	0.067	0.029
cycle_rate	0.000	0.027	0.136	0.000	0.000
kendall_zeta	0.557	0.912	1.000	1.000	1.000
contradiction_share	0.000	0.031	0.124	0.000	0.000